■ Project Documentation — English

AI Investment

Robot

How AI reads the news and analyzes stocks

to help you make smarter investment decisions

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■ Python · Llama 3.2 · NewsAPI · yfinance

■ This document is for educational purposes only. Not financial advice.

1 ■ What is the AI Investment Robot?

Imagine you want to invest in the US stock market, but you don't have the time — or the patience — to follow news, charts and financial reports all day long. That's exactly where the **AI Investment Robot** comes in: a program that does all of that for you and delivers a clear answer: **Buy (BUY)**, **Hold (HOLD)** or **Sell (SELL)**.

The project runs entirely on your local computer — no data is sent to external AI servers. All analysis is performed locally using the Llama 3.2 model via Ollama.

Why was this project created?

This project was born from a combination of three things: **genuine curiosity**, **hands-on learning** and a **passion for investing**.

I'm on a learning journey in Artificial Intelligence, and I believe the best way to truly learn any technology is to **build something real with it** — a project with a real purpose. Studying theory is important, but it's when you build from scratch that concepts actually click.

> ■ **The personal motivation:** Two subjects that have always fascinated me are **Artificial Intelligence** and the **financial market**. This project is exactly where those two worlds meet — a chance to combine technical learning with something I genuinely care about and that works in the real world.

The AI Investment Robot was also a way to implement in practice concepts such as **local language models (LLMs)**, **asynchronous processing**, **AI-powered sentiment analysis** and **multi-API integration** — all within a context that goes beyond a simple coding exercise: a system that delivers real value and can be used in the real world.

> ■ **The core idea behind the project:** News reflects market sentiment *before* it shows up in the stock price. If we can capture that sentiment in time, we have an informational edge.

The three sources of intelligence

■ Llama 3.2 (Local AI)	■ Linear Regression 30d	■ Alpha vs S&P 500
News Analysis	Price Trend	vs. Broad Market

Each of these three sources captures something different about the stock being analyzed. Combined, they form a single composite score that drives the final decision.

Which stocks does it analyze?

Only US companies listed on **NYSE** and **NASDAQ** — such as Apple (AAPL), Microsoft (MSFT), Tesla (TSLA), NVIDIA (NVDA), Amazon (AMZN) and similar. This is intentional: the news API returns richer and more consistent results in English for these companies, significantly improving sentiment analysis quality.

> ■■ **Important disclaimer:** This system is an educational and quantitative analysis project. It does NOT constitute financial advice. Always consult a qualified financial advisor before making investment decisions.

2 ■■ How Does It Work? The 7-Step Pipeline

When you run the robot, it goes through **7 sequential steps**. The output of each step feeds the next, like an intelligent production line:

1 **Interactive Menu**
You enter the ticker and company name

2 **News Collection**
NewsAPI fetches up to 20 relevant articles

3 **Sentiment Analysis**
Llama 3.2 classifies each article

4 **Financial Data**
yfinance downloads 30 days of price history

5 **Indicators**
Trend, Alpha and Volume calculated

6 **Decision Engine**
Weighted score → BUY / HOLD / SELL

7 **Final Report**
HTML, JSON, CSV and Log files generated

Why this pipeline design?

Separating the system into distinct steps offers three practical advantages: **(1)** each module can be tested and improved independently; **(2)** sentiment analysis can run in parallel with financial data collection, reducing total time; **(3)** intermediate output files (JSON, CSV) allow further analysis and backtesting without re-running everything.

3 ■ News Collection — Reading the Market's Mood

The first thing the robot does after you enter the ticker and company name is fetch recent news from **NewsAPI**. It collects up to **20 articles**, sorted by relevance, extracting the title and summary of each one.

Why 20 articles?

It's the ideal balance: enough for reliable statistical analysis, without compromising the AI's response time. Each article will be individually analyzed by Llama 3.2 in the next step.

Data cleaning before analysis

Rule	What it does	Why
Removed filter	Discards articles with "[Removed]" in the title	Content deleted by the source after publication — invalid for analys
Null conversion	Empty (None) fields become empty strings	Prevents technical errors when sending the payload to the AI mode
Zero valid articles	System logs a warning and defaults to HOLD	Safety: without enough data, the robot avoids issuing a directional

4 ■ The AI Reads the News — Sentiment Analysis with Llama 3.2

This is the heart of the project. Each collected article is sent to **Llama 3.2**, running locally via **Ollama**. The AI reads the article's title and summary and classifies the sentiment as **positive**, **negative** or **neutral** — along with a short natural-language justification.

Why use AI instead of traditional techniques?

> Traditional text analysis techniques (like VADER) fail to capture complex financial context. For example: *"Apple cuts sales forecast"* is unambiguously negative for an investor — but a purely lexical model may classify it as neutral, since it finds no explicitly negative words. Llama 3.2 understands the business context and reasons about implications, exactly as a human analyst would.

How the prompt is structured

The "prompt" is the instruction sent to the AI. It was carefully designed to ensure consistent, programmatically parseable responses:

```
System: You are a financial analyst specialized in market news sentiment analysis.
        Analyze the article about {company}.
        Respond ONLY in this format: <category> - <justification>
        Where <category> is: positive, negative or neutral
        Example: negative - regulatory scandal involving user data

User: Title: {article title}
        Summary: {article summary}
```

Design Decision	Reason
Strict format (category - justification)	Enables automatic parsing and reduces ambiguous responses
Temperature = 0.1 (near zero)	Maximizes consistency and determinism across runs
Example included in prompt (few-shot)	Anchors the model to the expected format without extensive examples
Financial analyst role in system prompt	The AI interprets news from an investor's perspective

What if the AI responds unexpectedly?

The system has a 3-step fallback mechanism: **(1)** tries to split on the first " **-** " in the response; **(2)** if that fails, searches for the keyword anywhere in the response; **(3)** if still unrecognized, classifies as "neutral" and logs the error — the pipeline is never interrupted by this.

Parallel processing — real speed

Instead of sending one article at a time (which would be very slow), the robot uses **asynchronous processing** to fire up to 3 simultaneous calls to Llama. Result: 20 articles processed in **30 to 90 seconds**, versus the 10 to 20 minutes it would take sequentially.

■ **How sentiment becomes a number:** Each classification receives a score: positive = +1 | neutral = 0 | negative = –1. The average across all articles is the **sentiment score** used in the decision engine.

5 ■ Financial Data — What the Numbers Say

While the AI reads the news, the robot also collects **30 days of price history** for the analyzed stock — plus the same period's data for the S&P; 500 index (the main barometer of the US market). All via **yfinance**, a free library that accesses Yahoo Finance data.

Why 30 days of history?

> It's the balance between relevance and statistical robustness. Very short windows (7 days) are unstable and noisy — a single bad day can distort everything. Windows that are too long (90+ days) may reflect market conditions that are no longer relevant to the current decision.

The three calculated indicators

1. Price Trend — Linear Regression

Instead of simply comparing the first and last day's price (which would be easily distorted by extreme volatility), the robot uses **linear regression** over 30 days. This gives a view of the overall trend, much more robust to outliers.

Metric	What it means	Example
slope_pct_per_day	Percentage price change per trading day	+0.23% per day → uptrend
r_squared	Trend consistency (0 = noise, 1 = perfect trend)	0.82 → well-defined trend
trend	Simplified trend direction	'upward' or 'downward'

2. Alpha vs S&P; 500 — Relative Performance

Alpha measures the stock's return in excess of the market's return over the same period. It's a valuable indicator because it **isolates the company's performance from the broad market's movement**. A stock can fall 2% but show positive alpha if the S&P; 500 fell 5% in the same period — that signals relative strength, not weakness.

Alpha	Interpretation
Alpha > 0	Stock outperformed the market — sign of relative strength → favors BUY
Alpha = 0	Stock tracked the market — neutral for the decision
Alpha < 0	Stock underperformed the market — sign of weakness → favors SELL

3. Volume Analysis — The Modifier

Volume (number of shares traded) doesn't indicate direction — but it amplifies or attenuates the other signals. The logic is simple: a price move backed by high volume is far more credible than the same move on low volume.

Volume Signal	Criterion	Effect on Final Score
■ High	Last day > 30d average by +20%	Score multiplied by 1.15 (+15%)
■ Normal	Variation between –20% and +20%	No change
■ Low	Last day < 30d average by –20%	Score multiplied by 0.85 (–15%)

6 ■■ The Decision Engine — How the Final Call is Made

Now that we have three independent signals — news sentiment, price trend and alpha vs market — the decision engine combines them into a single **composite score**, with different weights for each factor:

The score formula

```
Score = (Sentiment × 0.45) + (Trend × 0.35) + (Alpha × 0.20)

       If Volume high: Score × 1.15
       If Volume low: Score × 0.85
```

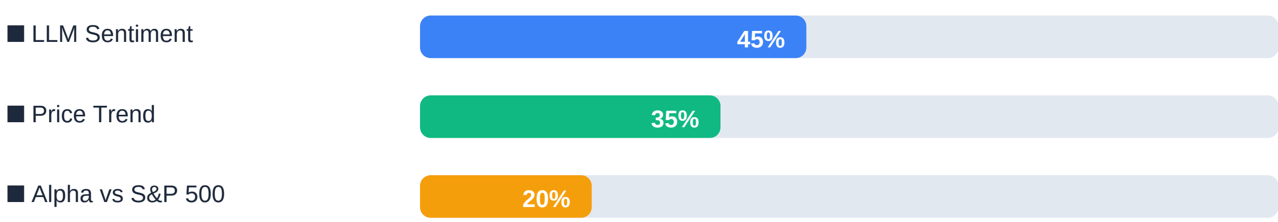
The final decision

✓ BUY
Score > 0.25
Positive signals dominate

■ HOLD
-0.25 ≤ Score ≤ 0.25
Mixed or weak signals

✗ SELL
Score < -0.25
Negative signals dominate

Why these weights?



Factor	Weight	Rationale
■ LLM Sentiment	45%	Captures qualitative information the market hasn't priced in yet. The system's differentiator.
■ Price Trend	35%	30-day momentum is reliable and based on objective data.
■ Alpha vs S&P 500	20%	Relative strength — lower weight as it can be influenced by sector-wide factors.
■ Volume (modifier)	±15%	No directionality — only amplifies or attenuates the other signals.

Adjusting the risk profile

The default thresholds (±0.25) are set to require **at least two of the three factors to point in the same direction** before issuing a directional signal.

- **Conservative profile:** increase to ±0.35 (fewer signals, safer)
- **Aggressive profile:** decrease to ±0.15 (more signals, higher risk)

7 ■ Generated Outputs — What the Robot Delivers

At the end of each analysis, the robot automatically generates **4 files** in the *outputs/* folder and maintains a persistent log in *logs/*:

File	Format	Contents
report_{TICKER}_{TS}	HTML	4 interactive charts, metric cards, news table, mathematical rationale for the decision
data_{TICKER}_{TS}	JSON	All analysis data: decision, indicators, news with scores, 30-day price history
news_{TICKER}_{TS}	CSV	News table: title, summary, sentiment, justification, score, source, date
prices_{TICKER}_{TS}	CSV	Price history: date, close, volume, cumulative percentage return
investment_robot	LOG	Persistent log with timestamps for each step — never overwritten between runs

What's in the HTML Report?

Section	What it shows
■ Decision Hero	Color-coded badge (green=BUY, red=SELL, amber=HOLD), confidence bar and score breakdown
■ Key Metrics	4 cards: last price, 30d return, alpha and average sentiment score
■ Sentiment Analysis	Positive/neutral/negative distribution + donut chart
■ Price History & Trend	Line chart with prices + regression line + R^2 and slope explanation
■ Cumulative Return vs S&P 500	Visual performance comparison — shows alpha graphically
■ Volume Analysis	Volume bars + average line + explanation of the modifier applied
■ Decision Rationale	Step-by-step table with each mathematical operation in the score
■ News Articles	Full table of all news articles with links, sentiment and AI justification

8 ■ How to Install and Run

What you need installed

Requirement	Details	Where to get it
Python 3.12	Recommended via Conda	anaconda.com
Ollama	Local AI server	ollama.com/download
Llama 3.2 (~4GB)	Local language model	Downloaded via Ollama
NewsAPI Key	Free tier works	newsapi.org

Step-by-step installation

1 **Create the Python environment**

```
conda create --name ai-robot python=3.12 conda activate ai-robot
```

2 **Install dependencies**

```
pip install numpy pandas scipy yfinance newsapi-python httpx python-dotenv
```

3 **Set your NewsAPI key**

```
echo NEWSAPI_KEY=your-key-here > .env
```

4 **Start the Ollama server**

```
ollama serve
```

5 **Run the robot!**

```
python projeto10-llama-fixed.py
```

■ After running, the system will ask for: **(1)** the stock ticker — e.g. AAPL, NVDA, TSLA — and **(2)** the company name for the news search — e.g. Apple, NVIDIA. Everything else happens automatically!

9 ■■ Configurable Parameters

All adjustable parameters sit at the top of the main file as constants. You can customize the robot's behavior without touching the core logic:

Parameter	Default	What it does
LLM_MODEL	"llama3.2"	Ollama model. Can be swapped for llama3.1, mistral, etc.
MAX_CONCURRENT	3	Simultaneous LLM calls. Increase if your hardware supports it.
NEWS_PAGE_SIZE	20	Number of articles fetched per analysis.
HISTORY_PERIOD	"1mo"	Historical period. Accepts: 1mo, 3mo, 6mo, 1y.
BENCHMARK_TICKER	"^GSPC"	Benchmark. ^GSPC = S&P 500. Can swap for ^NDX (Nasdaq).
BUY_THRESHOLD	0.25	Minimum score for BUY. Increase for conservative profile.
SELL_THRESHOLD	-0.25	Maximum score for SELL. More negative = more conservative.
OLLAMA_TIMEOUT	120	Timeout in seconds per call. Increase for larger models.

10 ■■ What's Next? Future Roadmap

The AI Investment Robot already delivers real value today — but there's plenty of room to grow. These are the extensions planned for future versions:

■■ **Brazilian Market (B3)**
Integrate Infomoney and Valor Economico news feeds to cover PETR4, VALE3, ITUB4.

■ **Multi-Asset Analysis**
Accept a list of tickers and generate a comparative ranking for portfolio selection.

■■ **Historical Backtesting**
Simulate the robot's decisions over 12 months of data and compare with buy-and-hold.

■ **Automated Scheduling**
Cron job for daily pre-market analysis with email or Telegram delivery.

■ **Real-Time Dashboard**
Web interface with Streamlit or FastAPI + React for live visualizations.

■ **Brokerage Integration**
Alpaca API (US) for automatic orders when the score crosses user-defined thresholds.

Multi-LLM Voting
Run each article through 2+ models and use majority voting for greater robustness.

Legal Disclaimer: This system is a quantitative analysis project developed for informational and educational purposes. It does NOT constitute financial advice, investment recommendation, or solicitation to buy or sell any security. Past performance is not indicative of future results. Always consult a qualified financial advisor before making investment decisions.